IOTAFLOW
SYSTEMS PVT. LTD

Model VNS

" if you want to manage it Measure it "

Compact Electromagnetic Flow Sensor

- Measurement of pulsating flow
- Measurement of Sodium Hypochlorite and Caustic Soda
- Detection of gas lock of a constant volume pump

This is the most suitable for water treatment facilities, etc.

Also usable for steady flow besides pulsating flow.

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Specifications Other special specification, please contact us.

Classification	Item		Details		
	Model		VNS05-F/E	VNS10-F/E	VNS20-F/E
Pulsation measurement mode specification	Accuracy assurance - flow rate range (L/min)		0.01~1	0.1~10	0.6~60
	Accuracy (Liquid temperature at 25°C)	Unit pulse	±5.0%RS (100 to 20% of maximum flow rate) ±1.0%FS (20 to 1% of maximum flow rate)		
Stationary flow measurement mode specification	Accuracy guarantee - flow rate range (L/min)		0.05~1	0.5~10	3.0~60
	Low flow cutoff (L/min)		0.025	0.25	1.5
	Accuracy (Liquid temperature at 25°C)	Frequency pulse	±2.5%RS (100 to 20% of maximum flow rate) ±0.5%FS (20 to 5% of maximum flow rate)		
		Unit pulse	±2.0%RS (100 to 20% of maximum flow rate) ±0.4%FS (20 to 5% of maximum flow rate)		
Common specification for each mode	Pipe installation		Socket end union joint 16A	Socket end union joint 20A	
	Fluid temperature range		0 to 40°C (No freezing)		
	Fluid conductivity rate		14mS/cm or more		
	Nature of object liquid		VNS□□-F: For sodium hypochlorite [Concentration 1 to 12%]		
			VNS□□-E: For caustic soda [Concentration 10 to 25%]		
	Working pressure		1 MPa or less		
	Pressure loss		0.02MPa or less		
	Ambient temperature/humidity		Temperature: -10°C to 60°C Humidity: 35% to 85%RH (No freezing)		
	Responsiveness		63% response Damping time: 2s (Standard)		
	Signal cable		Length: 0.5m 4core (Red: Power supply + wire / Blue: GND / White: Output 1 / Yellow: Output 2)		
	LED display		Two-color 1 light (Green: Blinks during measurement Red: light up or Blinks in the event of error)		
	Installation position		Free (Vertical piping is recommended)		
	Common specification of output		NPN Open collector Current capacity; 20mA or less Voltage: 30VDC or less Residual voltage at turning ON: 1 V or less		
	Output 1 *2	Frequency pulse *1 / Duty ratio: 50%	Standard 200Hz (Settable in steps of 0.1Hz between 20 and 400Hz)		
		Unit pulse	0.001L/P(Standard)	0.01L/P(Standard)	0.1L/P(Standard)
		Alarm *3	Selectable between Normal Open (Standard) and Normal Close Details of alarm (Low source voltage/Meter NG/Non water/Excessive flow rate/Excessive liquid noise/Reverse-flow)		
		Switch *4	Selectable between Normal Open (Standard) and Normal Close Level judgment value: Settable in steps of 1% from 0 to 100%. (Convert the maximum flow rate 100%)		
	Output 2 *2	Unit pulse	Same as output 1		
		Alarm *3	Same as output 1		
		Switch *4	Same as output 1		
	Protection level		IPX4 (Equivalent to IP64)		
	Consumption current		60mA or less		
	Power supply		24 VDC (± 10%) Power supply shell be supplied with isolated power supply and 1 unit of power supply is connected to one unit of VNS. In addition, in the case of pulsating flow mode measurement, grounding of power supply FG terminal (Class D or more) is required.		
	Wetted materials		VNS□□-F: Main body (PEEK resin) / Electrodes * earth ring (Titanium) / O-ring (FKM) VNS□□-E: Main body (PEEK resin) / Electrodes * earth ring (Equivalent to hastelloyC22) / O-ring (EPDM)		

*1: Frequency at maximum flow rate
*2: For the set value and choice of Output 1 and Output 2, all are factory setting, and setting change after installation is not available.
*3: Alarm is selectable only either one of Output 1 or Output 2.
*4: For the window judgment, Output 1 and Output 2 are used.
(Note) CE mark certification for lightening surge has not obtained.

Feature

Capable of detecting the gas lock of sodium hypochlorite

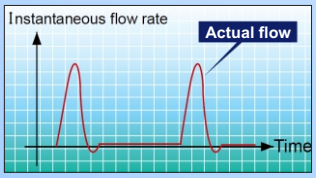
Gas lock detection is possible through the interlocking with actuating signal of pump. Gas lock is a phenomenon under which gas accumulates in a certain chamber of diaphragm of the metering pump and sodium hypochlorite is not transferred.

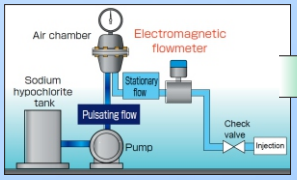
Air chamber is not required

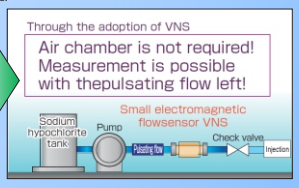
Is a measurement performed eliminating the pulsation with air chamber In the air chamber, time lag arises up to the buildup of pressure, therefore, a delay occurs in the measurement of flow rate.

Image of discharge flow from diaphragm electromagnetic metering pump

*Pulsating flow other than diaphragm electromagnetic metering pump is not covered by our accuracy assurance.







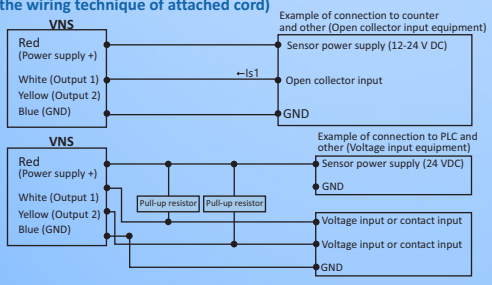
Wiring diagram (As to the wiring technique of attached cord)

Please set the sink current for open collector output of output 1 and output 2 to 20mA or less. For your reference, the recommendation for pull-up resistor becomes [10 to 20kΩ]

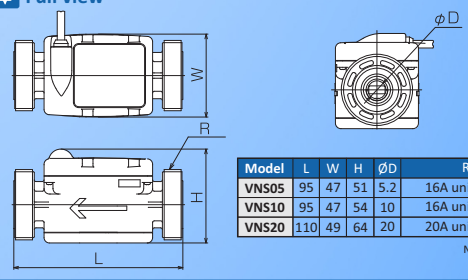
Sink current calculation formula

$$Is = \frac{V}{R} \leqq 20\ [mA]$$

V: Power supply (V)
R: Pull-up resistor (Ω)



Full view



Unit: mm

Model	L	W	H	ØD	R (Note)	Weight
VNS05	95	47	51	5.2	16A union (Male)	210g
VNS10	95	47	54	10	16A union (Male)	210g
VNS20	110	49	64	20	20A union (Male)	310g

Note: Exclusive of union
Cable length: 0.5m

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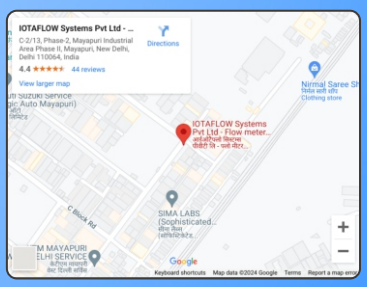

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